

Supplementary Material: OCV and Aging Models

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1 Supplementary OCV and Aging Models

This supplementary material provides the open-circuit-voltage (OCV) functions, baseline aging models, and parameter values used in the simulation study. These models are used only to evaluate the proposed control strategy in long-term simulations; they are not embedded in the controller. Thus, the proposed control strategy remains model-agnostic with respect to battery degradation.

1.1 OCV–SOC Relationships

Two lithium-ion chemistries are considered in the simulation study: lithium iron phosphate (LFP) and lithium manganese oxide (LMO). The LFP OCV–SOC relationship is adopted from [1]:

$$OCV_{\text{LFP}} = -0.5863 \exp(-21.9 \text{ SOC}) + 3.414 + 0.1102 \text{ SOC} - 0.1718 \exp\left(\frac{-0.008}{1 - \text{SOC}}\right), \quad (1)$$

where $\text{SOC} \in [0, 1]$. The LMO OCV–SOC relationship is adopted from [2]:

$$OCV_{\text{LMO}} = 3.875 - 0.335(-\ln(\text{SOC}))^{0.653} - 0.5332 \text{ SOC} + 0.8315 \exp(0.6(\text{SOC} - 1)), \quad (2)$$

where $\text{SOC} \in [0, 1]$.

1.2 Baseline LFP Aging Model

For LFP cells, the baseline short-term degradation model is adopted from [3]. The degradation over a short interval is modeled as

$$\Delta \text{SOH}_{\text{LFP}} = (aT^2 + bT + c) \exp((dT + e)I^{no}) \frac{|\Delta \text{SOC}|}{4.6} + \frac{f}{2.3} \exp(g\overline{\text{SOC}} + h/T) \frac{\Delta t}{2\sqrt{t}}. \quad (3)$$

Here T is the temperature in Kelvin, I^{no} is the average C-rate, $\overline{\text{SOC}}$ is the average SOC over the interval, ΔSOC is the SOC change over the interval, t is the accumulated calendar-aging time, and Δt is the duration of the current interval. During relaxation, $\Delta \text{SOC} = 0$ and $I^{no} = 0$, so (3) reduces to the calendar-aging term.

1.3 Baseline LMO Aging Model

For LMO cells, the baseline aging model is adopted from [4]. The degradation over a short interval is modeled as

$$\begin{aligned} \Delta \text{SOH}_{\text{LMO}} = & \left[\alpha_{\text{sei}} \beta_{\text{sei}} \exp(-\beta_{\text{sei}}(f_t + \sum_{i=1}^N f_{c,i})) \right. \\ & \left. + (1 - \alpha_{\text{sei}}) \exp(-(f_t + \sum_{i=1}^N f_{c,i})) \right] \left(f_{c,N+1} + \frac{f_t \Delta t}{t} \right), \end{aligned} \quad (4)$$

where N is the number of completed half cycles, and α_{sei} and β_{sei} are model parameters. The cycling-related term $f_{c,i}$ and calendar-aging term f_t are

$$f_{c,i} = \frac{0.5}{k_{\delta 1} |\Delta SOC_i|^{k_{\delta 2}} + k_{\delta 3}} \exp \left(k_{\sigma} (\overline{SOC}_i - \sigma_{\text{ref}}) + \frac{k_T T_{\text{ref}} (T - T_{\text{ref}})}{T} \right), \quad (5)$$

$$f_t = k_t t \exp \left(k_{\sigma} (\overline{SOC} - \sigma_{\text{ref}}) + \frac{k_T T_{\text{ref}} (T - T_{\text{ref}})}{T} \right). \quad (6)$$

Here ΔSOC_i and $\overline{SOC}_i$ denote the SOC swing and average SOC of the i th half cycle, respectively. For the current interval, $\Delta SOC_{N+1} = \Delta SOC$ and $\overline{SOC}_{N+1} = \overline{SOC}$.

1.4 Model Parameters

The parameters used in the LFP and LMO baseline aging models are listed in Table 1.

Table 1: Parameters of the baseline battery aging models.

Parameter	Value	Parameter	Value
a	2.0916×10^{-8}	b	-1.2179×10^{-5}
c	1.80×10^{-3}	d	-1.7082×10^{-6}
e	5.56×10^{-2}	f	5.9808×10^6
g	6.898×10^{-1}	h	-6.4647×10^3
α_{sei}	5.75×10^{-2}	β_{sei}	121
$k_{\delta 1}$	1.40×10^5	$k_{\delta 2}$	-5.01×10^{-1}
$k_{\delta 3}$	-1.23×10^5	k_{σ}	1.04
σ_{ref}	0.50	k_T	6.93×10^{-2}
T_{ref}	298.15 K	k_t	$4.14 \times 10^{-10} \text{ s}^{-1}$

1.5 Pack-Level Cell-to-Cell Heterogeneity and Knee-Point Acceleration

The baseline aging models in Sections 1.2 and 1.3 do not explicitly account for cell-to-cell heterogeneity within a pack. In addition, they were primarily calibrated in the early-life regime, whereas many lithium-ion cells exhibit accelerated degradation after a knee point. To represent these two effects in the pack-level simulations, the baseline degradation model is augmented with a cell-specific multiplicative factor and a knee-point acceleration term.

For either chemistry, the short-term degradation of cell i is computed as

$$\Delta SOH_i = \gamma_i [1 + 50 \max\{SOH_{\text{knee}} - SOH_i, 0\}] \Delta SOH_{\text{base}}, \quad (7)$$

where ΔSOH_{base} is the degradation predicted by the corresponding chemistry-specific baseline aging model. The multiplicative factor $\gamma_i \sim \mathcal{N}(1, \sigma_{\gamma}^2)$ is sampled once for each cell and held constant over time, representing persistent cell-to-cell variation in aging rate. Unless otherwise stated, $\sigma_{\gamma} = 0.1$, corresponding to a 10% RMS variation in aging rate under identical operating conditions, and $SOH_{\text{knee}} = 75\%$. The knee-point term increases the degradation rate once SOH_i drops below SOH_{knee} , thereby approximating late-life acceleration.

References

- [1] L. Lam, P. Bauer, and E. Kelder, “A practical circuit-based model for li-ion battery cells in electric vehicle applications,” in *2011 IEEE 33rd International Telecommunications Energy Conference (INTELEC)*. IEEE, 2011, pp. 1–9.
- [2] C. Zhang, J. Jiang, L. Zhang, S. Liu, L. Wang, and P. C. Loh, “A generalized soc-ocv model for lithium-ion batteries and the soc estimation for lnmco battery,” *Energies*, vol. 9, no. 11, p. 900, 2016.
- [3] J. Nájera, J. Arribas, R. de Castro, and C. Núñez, “Semi-empirical ageing model for lfp and nmc li-ion battery chemistries,” *Journal of Energy Storage*, vol. 72, p. 108016, 2023.
- [4] B. Xu, A. Oudalov, A. Ulbig, G. Andersson, and D. S. Kirschen, “Modeling of lithium-ion battery degradation for cell life assessment,” *IEEE Transactions on Smart Grid*, vol. 9, no. 2, pp. 1131–1140, 2016.